

Generalized Gross-Pitaevskii Equation for 2D Bosons with Attractive Interactions

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We introduce a generalized Gross-Pitaevskii equation that provides a nonlinear framework for studying 2D attractive Bose systems. Its defining feature is the logarithmic density dependence of the coupling constant, which breaks the scale invariance inherent in the standard mean-field equations. This framework allows straightforward calculations of the system properties arising from the quantum anomaly. As a first illustration, we study universal bound states in free space, commonly referred to as quantum droplets. Then, we analyze breathing modes and quench dynamics in trapped systems, paving the way for a systematic exploration of nonequilibrium phenomena in 2D attractive Bose systems. Finally, we predict the existence of universal excited states, including vortex configurations, which may be more accessible to experimental investigation than the ground state. Our results provide a robust theoretical foundation for studying both static and dynamical properties of finite systems, and offer guidance for the design of future experiments.

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Scale invariance of the attractive two-dimensional (2D) Gross-Pitaevskii equation (GPE) gives rise to unique phenomena such as Townes solitons [1], “strong” self-similar collapse [2,3], and interaction-independent breathing-mode frequencies in tightly trapped systems [4,5]. In realistic quantum systems of single-component cold bosons with short-range interactions, this symmetry is, however, broken by the presence of the two-body bound state that introduces an explicit length scale into the problem. One of the most striking consequences of this “quantum anomaly” [6,7] is the formation of universal many-body bound states [8–10]. For a finite number of particles, the balance between kinetic energy and short-range interactions confines these states to a finite size, thus justifying their designation as quantum droplets. While their microscopic nature is clearly distinct from that of quantum droplets in two-component or dipolar Bose gases [11–13], they likewise arise from quantum fluctuations.

Despite this exciting progress in our understanding of 2D attractive bosons, it remains unclear whether a simple, unified theoretical framework exists to analyze these

phenomena. Consequently, each new solution represents a state-of-the-art effort; see, for example, recent studies in Refs. [14–16], which rely on beyond-mean-field, quantum Monte Carlo and variational approaches, respectively. Here, to address this issue, we introduce a generalized GPE that incorporates a physical length scale via a density-dependent coupling, see “Framework” below. This equation enables a straightforward and unified description of both the static and dynamic properties of the system, including excited droplet states.

To illustrate the framework, we first demonstrate that it successfully describes the behavior of universal droplets [8,14], while preserving a simple structure suitable for intuitive analytical and numerical exploration. For example, in our theory, the finite droplet size emerges naturally because the coupling constant vanishes at high densities, thereby preventing collapse due to attractive interactions (see “Benchmark” below). We note in passing that, because the coupling vanishes in the high-density limit, 2D bosons may become a useful toy model for the renormalization of strong interactions in particle physics [10]. Our methods and results may therefore be useful for building intuition about quantum chromodynamics (QCD) systems, e.g., quark matter with vortices [17,18]. Indeed, QCD is asymptotically free [19,20] and thus naïvely scale invariant at high energies, but this scale invariance is anomalous and broken by dimensional transmutation. In finite density QCD, a density-dependent coupling constant enters naturally [21], although in practice, the chemical potential is often used instead of the density [22].

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In “Discussion,” we demonstrate the capability of the generalized GPE by examining the universal ground state in a trap, and discussing corresponding breathing dynamics. Finally, we summarize our findings and outline prospects for future studies of quench dynamics and universal excited states, such as vortices.

Framework: Energy functional—We study an ultradilute system of N 2D bosons of mass m . In our analysis focused on the limit $N \gg 1$, we rely on the following approximation for the energy

$$\frac{E_N[\psi]}{N} = \int d\mathbf{x} \psi^* \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial \mathbf{x}^2} + W(\mathbf{x}) + \frac{gN}{2} |\psi|^2 \right] \psi, \quad (1)$$

where $\psi = \psi(\mathbf{x})$ is a normalized function ($\|\psi\| = 1$) found by minimization of E_N ; W is the trapping potential. To facilitate a clear and consistent illustration of our results, we either consider a free system ($W = 0$) or employ a harmonic trap, $W(\mathbf{x}) = m\omega^2 \mathbf{x}^2/2$. In what follows, we use the units with $\hbar = m = \omega = 1$. Energy is then expressed in the units of $\hbar\omega$; interaction strength, g , in $\hbar^2/m$; length in $\sqrt{\hbar/(m\omega)}$; time and frequencies in $1/\omega$ and ω , respectively. We will also often express length in the units of the state size, $R = \sqrt{\int d\mathbf{x} |\mathbf{x}\psi|^2}$, which is the only relevant length scale for $W = 0$ or equivalently in the regime of strong attractions.

A unique aspect of 2D many-boson systems is that their theoretical description requires the coupling constant g to depend on the system’s probability density, $n = |\psi|^2$, even in the low-energy (i.e., weakly interacting) regime. For repulsive boson-boson interactions, $g > 0$, a widely used expression is $g = 4\pi/|\ln(Nna^2)|$ [23–25], where a denotes the 2D scattering length. The logarithmic dependence of g on n is notably weak, which poses challenges for experimental detection [26]. Nevertheless, this density dependence has a fundamental impact—it underlies the aforementioned symmetry breaking, as the scale invariance of the mean-field approximation relies on the assumption that g remains constant.

For attractive interactions and in the absence of an external potential ($W = 0$), a possible form of the coupling constant is $g \simeq -4\pi/|\ln(R^2 B_2)|$ [8], where $B_2 > 0$ denotes the two-body binding energy. This expression is, however, not as practically useful as the one for repulsive interactions. In particular, it cannot be straightforwardly extended to include an external trap—essential for modeling cold-atom systems—since the parameter R must then be determined from a separate equation, which is challenging to solve numerically [16]. Furthermore, it is unclear how to use this coupling constant in time-dependent problems. In what follows, we propose a modified form of g that circumvents these limitations.

Framework: Coupling constant—For $W = 0$, the bosonic system forms a droplet whose size, R^2 , is exponentially

smaller than that of the corresponding two-body bound state, with the exponent set by the particle number N [8]. Consequently, the coupling g in free space is mainly determined by the value of R^2 .

The size R is the only length scale of the many-body problem that determines the properties of the droplet, e.g., its density is given by $N|\psi|^2 \sim F(r/R)^2/R^2$, where $r = |\mathbf{x}|$; F denotes a universal function that captures the shape of the droplet. The exponential dependence of the parameter R on N and the fact that the amplitude of the function ψ is given by $1/R$ motivates the following ansatz for g in the limit of large particle numbers, $N \gg 1$,

$$g \simeq -\frac{4\pi}{\ln[\alpha|\psi(\mathbf{x})|^2/B_2]}, \quad (2)$$

where $\alpha = 2.607$. Our choice of the parameter α is fixed by the ground state energy derived in Ref. [14], as described in the End Matter. For $W = 0$, Eq. (2) is (within logarithmic accuracy) equivalent to $-4\pi/|\ln(R^2 B_2)|$ derived in Ref. [8] everywhere except the points where $\psi(\mathbf{x})$ vanishes. This region, however, does not contribute to the energy functional in Eq. (1) where g is multiplied by $|\psi|^2$.

Although Eq. (2) is inspired by the strongly interacting regime ($B_2/\hbar\omega \gg 1$) where the trap can be neglected, it remains a natural choice for weakly interacting systems in a trap, i.e., when $B_2 \rightarrow 0$. Indeed, in this case, the coupling constant obeys $g \simeq -4\pi/|\ln(a^2)|$, see, e.g., Ref. [15]. Within logarithmic accuracy, this expression coincides with Eq. (2) (recall that $B_2 \sim 1/a^2$, see, e.g., Ref. [27]). Finally, to assess the validity of Eq. (2) at intermediate interaction strengths, we perform numerical calculations as presented in Fig. 1, see the discussion below and Supplemental Material [28]. [Intermediate interaction strengths correspond to $(\ln B_2)/N \simeq -2.15$ [44], see Refs. [15,16] and below.]

Framework: Gross-Pitaevskii equation—Once the coupling constant is established, we use Eq. (2) in Eq. (1) and the equation of motion $i\partial\psi/\partial t = \delta E[\psi]/\delta\psi^*$ to derive the generalized Gross-Pitaevskii equation that governs the physics of the problem

$$i\frac{\partial\psi}{\partial t} = \left[-\frac{1}{2} \frac{\partial^2}{\partial \mathbf{x}^2} + W(\mathbf{x}) + GN|\psi|^2 \right] \psi, \quad (3)$$

where $G = g + g^2/(8\pi)$ [46]. Equation (3) provides a powerful framework that greatly facilitates the analysis of 2D bosons with attractive interactions. To illustrate this, we start by calculating static properties of the system, i.e., we solve Eq. (3) in polar coordinates using the ansatz $\psi(\mathbf{x}, t) = e^{-i\mu t} e^{is\phi} \sqrt{n(r)}$, where μ is the chemical potential, and the topological quantum number s determines the vorticity of the system. This ansatz allows us to recover established results and enables new advances.

Benchmark: Strong interactions—For sufficiently strong interactions, the trap can be neglected [15,16], and the system forms a universal many-body bound state. Given a

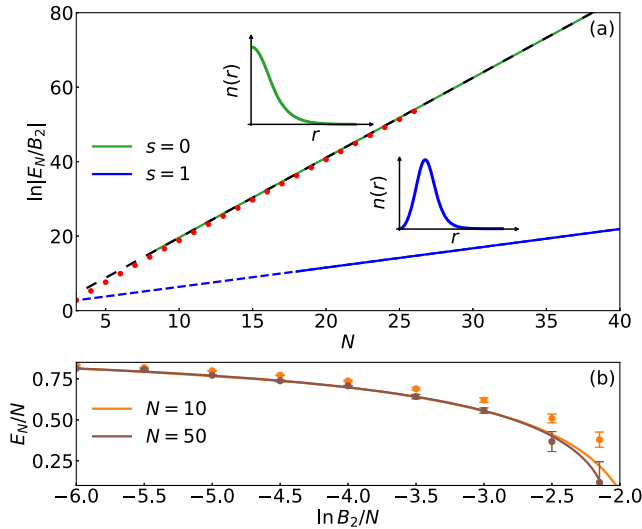


FIG. 1. (a) Energy, $\ln|E_N/B_2|$, as a function of the number of bosons N for the lowest energy states without ($s = 0$) and with ($s = 1$) vorticity for the untrapped system ($W \rightarrow 0$). The solid green line shows numerical solution of Eq. (3) for $s = 0$; the black dashed line shows the expected asymptote $\ln|E_N/B_2| = (4\pi N/\mathcal{G}) - 1.91$ [14]. The red dots demonstrate the numerically exact solution of the many-body problem available for “small” values of N [45]. The blue line shows the lowest energies for $s = 1$. The blue dashed line is a linear fit to this result, $\ln|E_N/B_2| = 0.5174N + 1.218$. The insets sketch the radial densities for the states. (b) The lowest energy solution of Eq. (3) in a trap for weak and intermediate interactions (solid lines) together with the numerical solution of the many-body problem obtained using flow equations for bosons (dots with error bars), see End Matter for details.

“weak” (logarithmic) dependence of G on $\psi(\mathbf{x})$, we gain analytical insight into the equilibrium properties of these states by treating G as constant. Under this assumption, Eq. (3) with $W = 0$ is stable only for specific values of $GN = \mathcal{G} < 0$, where $\mathcal{G}$ is independent of N [48]. These values correspond to the fixed points of the energy functional in Eq. (1), where the kinetic energy is balanced by particle-particle attraction. The corresponding solution ψ has the form $\psi \sim e^{is\phi} F_{\mathcal{G},s}(r/R)/R$, here $F_{\mathcal{G},s}$ is a universal function parametrized by $\mathcal{G}$ and s . The associated length scale follows from Eq. (2) for $N \gg 1$: $R^2 \simeq 1/B_2 e^{4\pi N/\mathcal{G}}$.

The smallest possible $|\mathcal{G}|$, $\mathcal{G} \simeq -5.84$, determines the ground state, whose static properties are by now well established [8,14]. Within our framework, these properties can be readily studied numerically using Eq. (3) [49], see Fig. 1(a). In particular, it is straightforward to recover the universal dependence of energy on the number of particles $E_N \simeq B_2 e^{4\pi N/|\mathcal{G}|}$ [8].

Among the most promising experimental platforms for observing this universality are trapped ultracold atomic gases [50]. To guide the corresponding experimental effort, the influence of the trapping potential on the system must be systematically explored. While this investigation is

numerically demanding and typically requires advanced computational methods [15], we demonstrate below that the generalized Gross-Pitaevskii equation enables accurate and computationally efficient analysis of these effects.

Benchmark: Intermediate interactions—To demonstrate the accuracy of the Gross-Pitaevskii equation in accounting for trap effects, we calculate the ground-state energy of the trapped system at intermediate interactions, i.e., when $|GN|$ is smaller than $|\mathcal{G}|$, which leads to $\ln B_2/N \lesssim 4\pi/\mathcal{G} \simeq -2.15$. This ensures the most pronounced interplay between trapping and interparticle interactions, cf. Ref. [16]. To perform calculations, we employ the flow-equation approach [51,52] (see also End Matter) in the specific form developed for degenerate Bose systems [28,53,54], which not only provide benchmark data but also represent the first application of these equations to two-dimensional systems, thereby opening new avenues for their extension to three dimensions.

The overall agreement between the data in Fig. 1(b) is good (see Supplemental Material [28] for the corresponding densities), validating the applicability of the generalized GPE in this regime. Note that the agreement between the results is better for $N = 50$ than for $N = 10$ (see Supplemental Material [28] for similar results for other particle numbers). This behavior, which mirrors the strongly interacting limit—see the difference between the red dots and the solid green line in Fig. 1(a)—is expected. Our parametrization of the interaction strength, particularly the numerical value of α , explicitly relies on the assumption that $N \gg 1$ (see End Matter). Nevertheless, the discrepancy is already marginal at $N \sim 10$ and does not affect our main conclusions. Having benchmarked the generalized Gross-Pitaevskii equation against established static results, we now apply it to explore less charted regimes. We begin by investigating trap effects in a regime beyond the reach of the flow equation method.

Discussion: Effect of the trap on universal states—To further investigate the effect of the trap, we solve Eq. (3) numerically (see Supplemental Material [28] for details). In the weakly interacting regime ($B_2 \rightarrow 0$), the density is predominantly shaped by the trapping potential. In contrast, for strong interactions ($B_2 \rightarrow \infty$), the universal (so-called Townes soliton [1]) profile is expected to emerge. Figure 2(a) illustrates the transition between these two limits. The rescaled density exhibits a crossover behavior, with only a weak dependence on the interaction strength [55].

It is important to note that the generalized Gross-Pitaevskii equation admits solutions for all finite values of B_2 and N , in contrast to the standard GPE, which exhibits mass concentration (collapse) as the interaction strength approaches a critical value from below [56]. Mass concentration cannot occur in our “asymptotically free” theory, because the coupling constant vanishes at high densities. In other words, the physics of the high-density regime is governed primarily by kinetic energy, which does not allow the system to collapse.

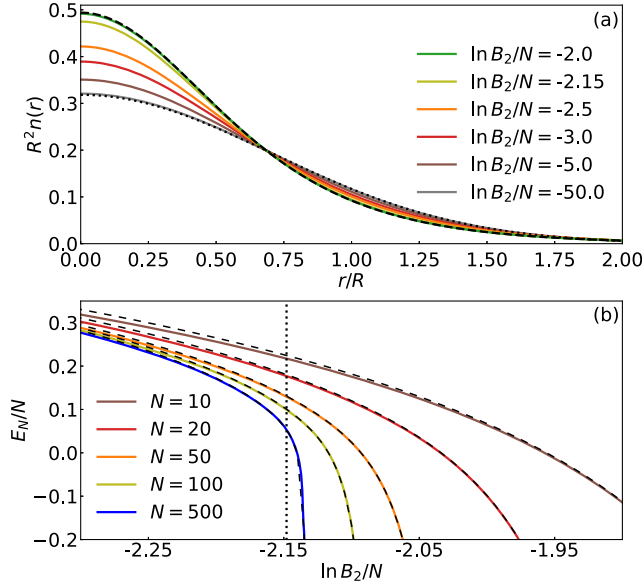


FIG. 2. (a) Density of the trapped Bose gas in the transition between the weakly and strongly interacting limits. The dimensionless density, $R^2 n(r)$, for the ground state without vorticity ($s = 0$) and $N = 50$ is plotted as a function of r/R for different interaction strengths $\ln B_2/N$. The dotted black curve shows the noninteracting harmonic oscillator solution, whereas the dashed curve represents the Townes-soliton profile. (b) Energy of the Bose gas per particle, E_N/N , as a function of the interaction parameter $\ln B_2/N$. The solid curves show the numerical solution to Eq. (3), the dashed black curves present the result of minimization of Eq. (1) with respect to the Townes-soliton profile [28]. Different curves correspond to different numbers of particles. The vertical dotted line shows the transition point in the limit $N \rightarrow \infty$ [15,16].

Instead of the collapse, in the generalized GPE, the energy shows a smooth increase, which becomes rapid near the critical parameter value $\ln B_2/N \simeq -2.15$, as shown in Fig. 2(b). The derivative of the energy also varies smoothly (see Supplemental Material [28]), suggesting a crossover rather than a sharp phase transition. Within our framework, the crossover also manifests in other system properties, such as the breathing frequencies, Ω .

Discussion: Breathing dynamics—To perform a numerical analysis of the breathing dynamics, we consider the time evolution of the system after a sudden change of the interaction from $\ln B_2/N$ to $\ln B_2/N - 0.001$. The resulting frequency is presented in Fig. 3. We see the crossover in the vicinity of $\ln B_2/N \simeq -2.15$ where the frequency of oscillations starts to rapidly increase. This increase becomes sharper as the number of particles increases, reflecting the results presented in Fig. 2(b).

The generalized Gross-Pitaevskii equation enables an analytical calculation of the frequency (see End Matter), thereby providing insight into the system’s limiting behavior. For weak interactions, we derive $\Omega \simeq 2 + N/(\ln B_2)^2$ [57]. The deviation of the breathing frequency from $\Omega = 2$,

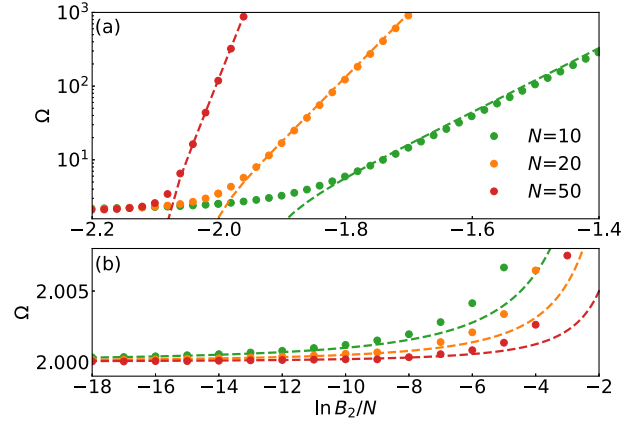


FIG. 3. Breathing frequencies Ω of a weakly perturbed trapped Bose gas as a function of the interaction strength $\ln B_2/N$ for different numbers of bosons N in (a) strong and (b) weak interaction regimes. Dots demonstrate frequencies from the numerical solutions. The dashed curves show analytical results valid for $N \gg 1$: (a) $3.8|E_N|/\sqrt{N}$; (b) $2 + N/(\ln B_2)^2$.

characteristic of the $SO(2,1)$ symmetry that arises from scale invariance in two dimensions [5], is a manifestation of “quantum anomaly” captured in our framework via a density-dependent coupling constant. For strong interactions, we derive a trap-independent frequency $\Omega \simeq 3.8|E_N|/\sqrt{N}$. This expression is in excellent agreement with recent free-space calculations by Petrov [14], further validating the generalized GPE.

Summary—In this Letter, we introduced and explored a nonlinear equation for attractive 2D Bose systems, providing a bridge to the well-established mean-field description of repulsive interactions [25,47,59]. The generalized GPE enables a quantitative description of the system using simple (semi)analytical tools. For instance, in the challenging regime of intermediate interaction strengths, the system’s energy can be obtained variationally by employing the Townes-soliton profile $f_T(r/\tilde{R})$ where $\tilde{R}$ is a variational parameter that minimizes the mean-field energy [see Fig. 2(b) and Supplemental Material [28]].

Outlook: Quench dynamics—We anticipate that one of the main applications of the generalized Gross-Pitaevskii equation will be the exploration of nonequilibrium dynamics and quantum control (see, e.g., Ref. [60] for a recent use of an effective mean-field approach in this context)—an area that appears otherwise inaccessible for 2D attractive Bose gases. To illustrate this possibility, we consider the quench dynamics of the system, a standard paradigm for investigating nonequilibrium physics in closed systems [61]. We initiate the non-interacting system in the ground state of the harmonic oscillator at $t = 0$ and consider the time dynamics of $\langle r^2 \rangle$ at $t > 0$, assuming a finite value of B_2 , see Fig. 4. We see that for weak interactions, the quench dynamics generates oscillations at frequencies $\Omega \simeq 2$, consistent with the scale-invariant dynamics. At stronger interactions, this

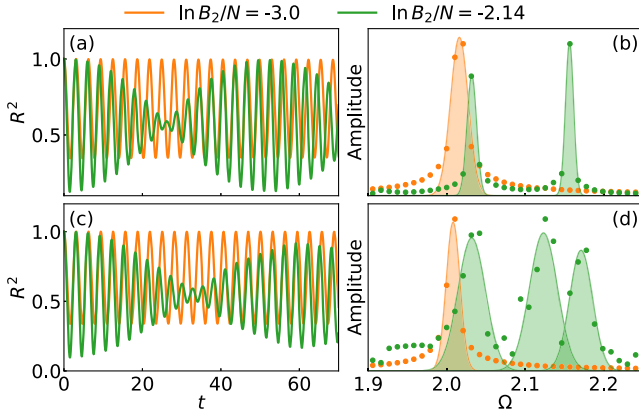


FIG. 4. Quench dynamics for a trapped Bose gas in the vicinity of the critical interaction strength with $N = 20$ (a),(b) and 50 (c),(d) from a noninteracting system to the interaction strength $\ln B_2/N = -3.0$ and -2.14 . Panels (a),(c) demonstrate the size of the system, R^2 , as a function of time t . Panels (b),(d) demonstrate the Fourier transforms of the R^2 signal (dots). To facilitate interpretation, we fit each peak with a Gaussian function (solid lines with shaded areas).

universality breaks down, and we observe an interference pattern (beating) in the time evolution, see End Matter.

Outlook: Universal excited states—Our work motivates further mathematical and numerical studies to validate the proposed framework and identify experimental conditions for observing 2D universality in attractive systems. A concrete direction involves studying universal excited states predicted by the generalized GPE [62], which are connected to the spectrum of scale-invariant solutions to the standard 2D GPE, either without [66] or with vorticity [67], see Ref. [48] for a review.

To illustrate one such state, we considered the vortex solution. Specifically, Fig. 1(a) presents the energy of the lowest-energy configuration with topological charge $s = 1$. This solution corresponds to a critical interaction strength of $\mathcal{G} \simeq -24.15$. The energies of the corresponding states are $E_N^e \sim 1/R^2$; they lead to the universal scaling relation $E_{N+1}^e/E_N^e \simeq 1.7$, which may be contrasted with the ground-state scaling $E_{N+1}^g/E_N^g \simeq 8.6$ [8]. Note the markedly slower increase in energy with growing particle number—a distinctive trait of the excited-state spectrum. This suggests that topologically protected excited states may be more suitable for the experimental observation of 2D universality than the ground state. It is important to note that the stability of such vortex states (both in free space and in trapped systems) is an open question with direct relevance to an experiment. We are currently addressing this issue through both the Bogoliubov–de Gennes approach and dynamical perturbation tests.

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Data availability—The data that support the findings of this article are openly available [68]. The software enabling the reproduction of the data is openly available [69].

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End Matter

Parameter α —To determine α in Eq. (2) from the main text, we first compute the energies of the system in the limit $W \rightarrow 0$, where according to Ref. [14] the ground state energy for $N \gg 1$ approaches $B_N^{s=0} \rightarrow B_2 e^{(4N/C)+c_1}$ with $C \simeq 1.862$ and $c_1 \simeq -1.91$. By trial and error, we found that $\alpha \simeq 2.607$ reproduces this limiting behavior well, see Fig. 5(b). Other values of α can be used if one finds this necessary. In particular, our model does not rule out the possibility for N -dependent α : $\alpha(N) = \alpha_0 + \alpha_1 N^\gamma + \dots$. The inclusion of this N dependence might be needed to describe the energy in the few-body limit accurately. Below, we outline the basic intuition behind determining the value of α that can be used for a further exploration of the generalized GPE.

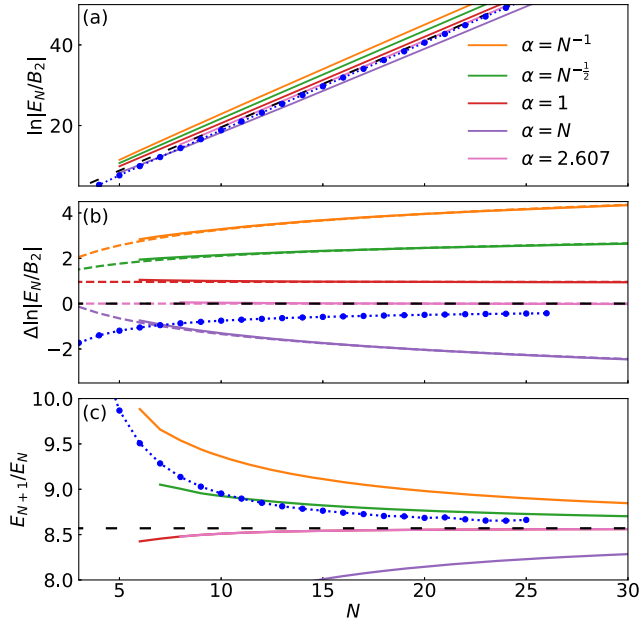


FIG. 5. (a) Ground-state energy of the untrapped Bose gas $\ln|E_N/B_2|$. (b) Energy differences between $\ln|E_N/B_2|$ and the expected asymptote $\Delta \ln|E_N/B_2| = \ln|E_N/B_2| - (4N/C + c_1)$ [14] for the data presented in panel (a). (c) Energy ratios E_{N+1}/E_N for the data presented in panel (a). Solid lines represent solutions of the generalized GPE. The black dashed line represents the solution of Ref. [14], and the blue dots with a dotted line show the results of Ref. [45]. In panel (b), the dashed colored lines show $\Delta \ln|E_N/B_2|$ where E_N/B_2 is calculated from Eq. (A3) with $\kappa = 1.448$.

In the limit $W \rightarrow 0$, $N \rightarrow \infty$, the many-body solution is well represented by the Townes soliton $\psi_T(r) = (2\pi C \mathcal{R}^2)^{-1/2} f_T(r/\mathcal{R})$, where $\mathcal{R}^2 = (C/M_2) \langle \psi_T | r^2 | \psi_T \rangle$ with $M_2 \simeq 2.211$ [70]. For an illustration of f_T , see Supplemental Material [28]. Therefore, the interaction strength in this limit should have the form

$$g = -\frac{4\pi}{\ln\left(\frac{\alpha(N)}{2\pi C} |f_T|^2 \frac{1}{\mathcal{R}^2 B_2}\right)}. \quad (\text{A1})$$

We rewrite this expression at the fixed point of the generalized GPE (note that $G \simeq g$ in the considered limit),

$$\mathcal{G} \simeq -\frac{4\pi N}{\ln\left(\frac{\alpha(N)}{2\pi C} F_{\text{eff}} \frac{1}{\mathcal{R}^2 B_2}\right)}, \quad (\text{A2})$$

where $\mathcal{G} = -\pi C$ and F_{eff} is a numerical constant representing some effective value of the density. The system's energy is $E_N = -(C/8\mathcal{R}^2)$ (see, e.g., Ref. [14]) and therefore

$$\frac{E_N}{B_2} = -\frac{\pi C^2}{4} \frac{1}{\alpha(N)} \frac{1}{F_{\text{eff}}} e^{\frac{4N}{C}}. \quad (\text{A3})$$

Now, we propose that $F_{\text{eff}} = \kappa |f_T(0)|^2$ with scaling coefficient κ .

To determine κ and illustrate the effect of possible dependence of α on N , we solve Eq. (3) numerically, for $\alpha(N) = N^\gamma$ with $\gamma = -1, 0, \frac{1}{2}, 1$. We use calculated values of E_N/B_2 and Eq. (A3) to fit the parameter $\kappa = 1.448$. Note that the parameter κ is independent of the $\alpha(N)$. In panels (a) and (c) of Fig. 5, we show $\ln|E_N/B_2|$ and E_{N+1}/E_N for different $\alpha(N)$ compared to Refs. [14,45]. Panel (b) presents the difference $\Delta \ln|E_N/B_2| = \ln|E_N/B_2| - (4N/C + c_1)$ where E_N/B_2 is calculated from the generalized GPE using Eq. (A3) with $\kappa = 1.448$.

Once κ is determined, we can estimate the parameter $\alpha(N)$ as

$$\alpha(N) = -\frac{\pi C^2}{4} \frac{1}{\kappa |f_T(0)|^2} \frac{B_2}{E_N} e^{\frac{4N}{C}}. \quad (\text{A4})$$

According to Ref. [14], $E_N = -B_2 e^{(4N/C)+c_1}$. Then

$$\alpha(N) = \alpha = \frac{\pi C^2}{4\kappa |f_T(0)|^2} e^{-c_1} \simeq 2.6073, \quad (\text{A5})$$

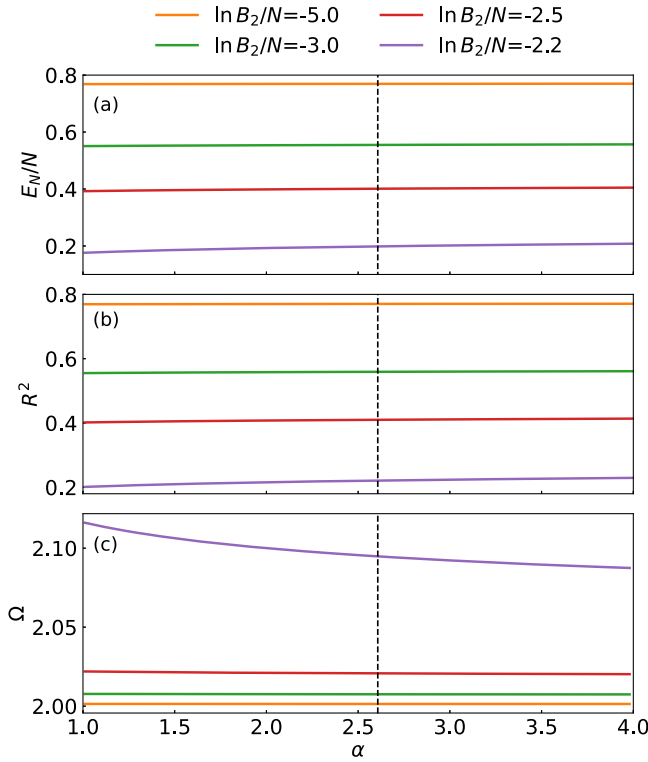


FIG. 6. (a) Ground-state energy E_N (b) the size of the system R^2 and (c) the breathing mode frequency Ω as a function of the parameter α for different interaction strength $\ln B_2/N$ for $N = 50$. Black vertical dashed line shows value of $\alpha = 2.6073$.

where we have used $f_T(0) = 2.207$, see, e.g., Ref. [8]. Results for $\alpha = 2.6073$ are also presented in Fig. 5. In Fig. 6, we show how the small variation of the parameter α affects energy E_N , the system's size R^2 , and breathing mode frequency Ω . In the weakly interacting regime, the properties are almost independent of α . The impact becomes more significant near the transition to the strongly interacting regime, where small variations of α can strongly affect the results.

In a more general approach,

$$E_N = -B_2 e^{\frac{4N}{C} + \theta(N)}, \quad (\text{A6})$$

where $\theta(N) = \sum_{i=1} c_i N^{\gamma_i}$. Parameters c_i and γ_i may be fitted to the results of other methods, see Ref. [45] for an example of $\theta(N)$ fitted to the few-body limit results. Equation (A6) leads to the expression

$$\alpha(N) = \frac{\pi C^2}{4\kappa |f_T(0)|^2} e^{-\theta(N)}, \quad (\text{A7})$$

where properly fitted $\theta(N)$ may improve the accuracy of generalized GPE in the few-body limit.

Flow equations—In the flow-equation method, we block diagonalize the Hamiltonian H using the flow equation

$$\frac{dH}{ds} = [\eta, H], \quad (\text{B1})$$

where s is the flow parameter (which can also be understood as an imaginary time). With the proper choice of the generator of the flow η , the off-diagonal matrix elements will vanish in the $s \rightarrow \infty$ limit. The Hamiltonian of the system reads

$$H = \sum_{i=1}^N \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial \mathbf{x}_i^2} + W(\mathbf{x}_i) \right) + \frac{1}{2} \sum_{i,j=1}^N V(\mathbf{x}_i - \mathbf{x}_j), \quad (\text{B2})$$

where V describes short-range particle-particle interactions; $\{\mathbf{x}_i\}$ are the coordinates of the bosons. We provide additional details in Supplemental Material [28].

Breathing dynamics—To calculate breathing-mode frequencies, we first compute the time dynamics of $R^2 = \langle r^2 \rangle$ (see Supplemental Material [28] for detailed derivation). The corresponding equation for large values of N is given by

$$\frac{d^2 R^2}{dt^2} = 4 \frac{E_N}{N} - 4R^2 + 2N \left\langle n^2 \frac{dg}{dn} \right\rangle. \quad (\text{C1})$$

This equation is general in the sense that it applies to any time dynamics of the state. For example, a time-independent problem requires $E_N = NR^2 - (N^2/2) \langle n^2 (dg/dn) \rangle$. Note that the total energy comprises two contributions: one from the harmonic confinement and another from density-dependent interactions. These contributions are balanced when $E_N = 0$, offering an intuitive explanation for the transition point between weakly and strongly interacting regimes, as illustrated in Fig. 2(b).

If g were a constant, the last term in Eq. (C1) would disappear, resulting in interaction-independent dynamics governed solely by the first two terms [4,5]. A density-dependent g breaks this picture, leading to a “quantum anomaly,” similar to dimensional transmutation in QCD [19,20] or the Efimov effect [71,72]. This symmetry breaking is an intrinsic property of two-dimensional systems with short-range interactions, contrasting with the extrinsic symmetry breaking induced by a “compactified” third dimension in cold-atom experiments [58].

We investigate “quantum anomaly” in the breathing dynamics, assuming that the solution has the form $\psi[r/R(t)]/R(t)$. The time dynamics of $R(t)$ follows from Eq. (C1): $\ddot{R}(t) = 4E_N/N - 4R^2(t) + N \overline{\langle n g^2 / 2\pi \rangle} / R^2(t)$, where the overline highlights that the expectation value is taken with respect to a rescaled function $\psi(r)$. Looking for small oscillations of $R(t)$ in the vicinity of the equilibrium value $R(t=0)$, we find the frequency of the breathing mode ($N \gg 1$)

$$\Omega = \sqrt{8 - 4E_N/(NR^2(t=0))}, \quad (\text{C2})$$

where we have used the time-independent solution to Eq. (C1). This expression agrees with the sum-rule approach result, see Supplemental Material [28].

Equation (C1) can also be used to analyze quench dynamics presented in Fig. 4. In particular, we can conclude that the beating in panel (c) arises from the time dynamics of g in Eq. (C1), which resonates with the universal, scale-invariant dynamics.